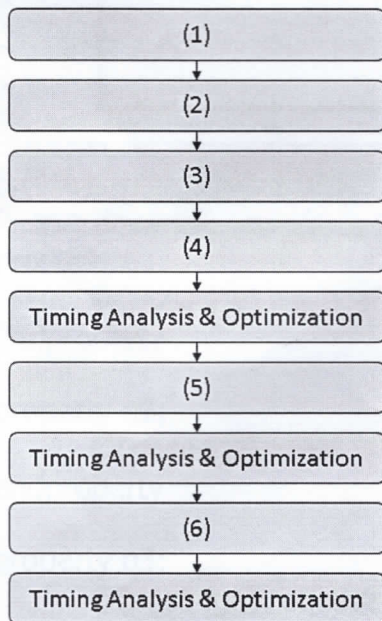


2016 ICLAB Final Exam

1. Please finish the APR flow as below. (12%)

(Example: (1)a (2)b (3)c (4)d (5)e (6)f)

- a. Standard Cell Placement
- b. Specify floorplan
- c. Clock Tree Synthesis
- d. Power Planning
- e. SI Driven Route
- f. IO, P/G Placement



2.

(a) What is antenna effect? (3%)

(b) How to avoid antenna effect? (4%)

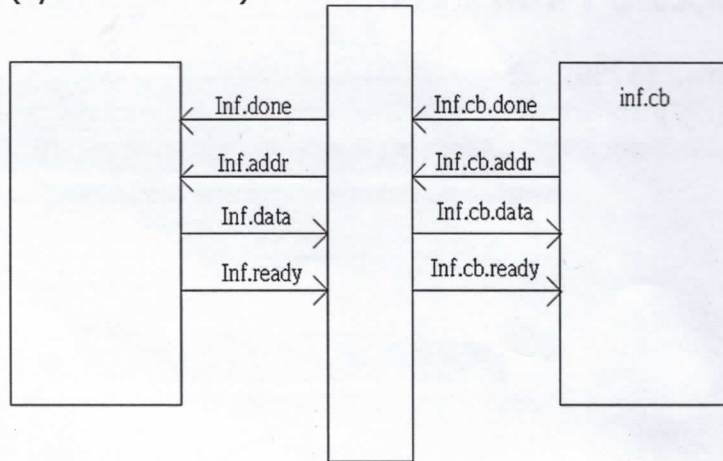
(c) Will antenna effect cause failure at post layout simulation? (Yes/No) (3%)

3. (10%)

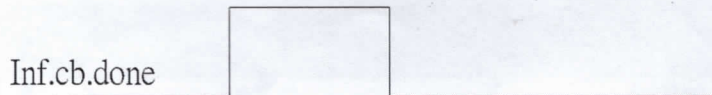
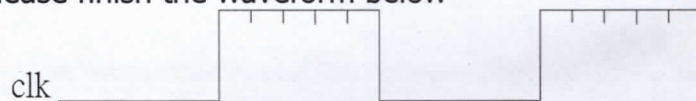
```
interface INF(input blk)
    logic data, ready, done, addr;
    clocking cb @(posedge clk)
        default input #2ns output #1ns
        input data , ready;
        output done;
        output #3ns addr;
    endclocking

    modport cb(clocking cb);
endinterface
```

(cycle time 10ns)



Please finish the waveform below



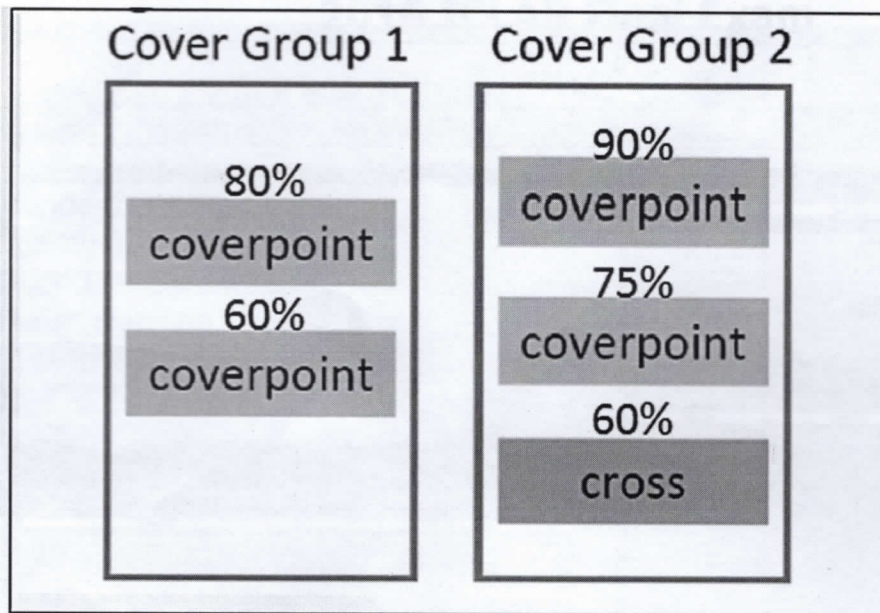
Inf.done



Inf.addr

4. (10%)

Calculate the coverage of the whole design, the coverage of coverpoints or cross is specified as shown on the picture



5.(9%)

Please draw the waveform to explain the following two assertions
property q1;

```
@(posedge) mem_en |-> (req ##3 ack);
endproperty
```

property q2;

```
@(posedge) mem_en |=> (req ##3 ack);
endproperty
```

property q3;

```
@(posedge) request ## 1 grant ##1 !request ##1 !grant;
endproperty
```

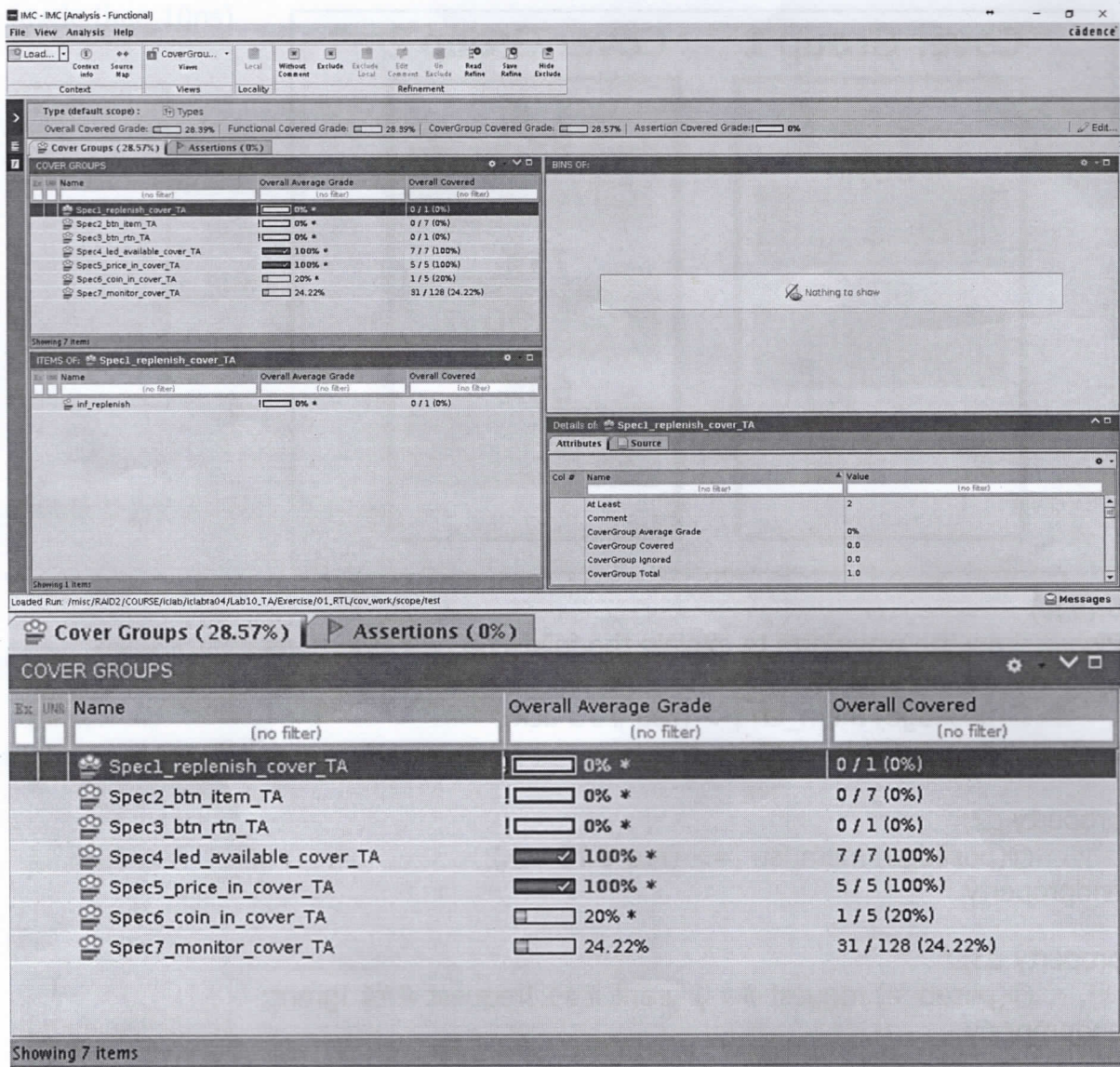
6.(10%)

Please simply describe how to let every component 100%.

$$\frac{0.05}{2} = 0.025$$

0.35

$$170 + 115 = 285$$



7. Please list 3 main reasons for chip to fail due to IR drop (10%)
8. What's information are in the technology ^{LEF}LDF file and Cell library LEF file?(10%)
9. please list four ways to reduce the power consumption, and briefly explain why.(10%)
10. Why do we proceed clock gating? Compare the AND-gating with the OR-gating, which one will consume less power?(10%)